MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
REGULAR EXAMINATION March 2026

Fourth SEMESTER

Skill Enhancement Course (SEC) - MG4SECFSC200 - GOOD LABORATORY PRACTICES

(2024 ADMISSION ONWARDS)

Duration:1 hour 30 minutes Maximum Marks: 50

PART A – Short Answers

1. Key points (any two):
   Accreditation body ensuring competence of laboratories
   Ensures quality, reliability, technical competence
   Follows ISO/IEC 17025 guidelines

2. Key points (any two):
   Ensures accuracy and precision
   Reduces measurement errors
   Ensures reliability of test results

3. Examples:
   Chemical/solvent fires
   Electrical fires, Fire during equipment malfunction

4. Key steps (any two):
   Use of standard buffer solutions (pH 4, 7, 10)
   Rinsing electrode before measurement
   Adjusting meter to match buffer value

5. International Organization for Standardization

6. Key points (any two):
   High fire hazard
   Requires ventilation/storage cabinets
   Use risk matrix (likelihood × severity)

7. Key points:
   Practical skill evaluation
   Written/Oral tests
   Observational assessments

8. Key points:
   Ensures consistency and accuracy
   Prevents errors and deviations

9. Key examples: Controlled access areas, Ventilation/fume hoods, Calibration laboratories, Proper storage areas

10. Key points: Fire hazard, Health risks due to fumes, Degradation of chemical quality

11. Key points: Tracks repairs and maintenance, Helps identify recurring problems, Ensures equipment uptime and quality

12. Examples: Lab coats, Gloves, Goggles, Aprons

PART B – Short Essay

4 marks: complete answer with examples
3 marks: minor omissions
2 marks: half answer
1 mark: minimal information

13. **Corrective action in quality management**
    Points:
    Identifies root cause of nonconformities
    Prevents recurrence
    Improves processes and performance
    Enhances reliability of results

14. Points
    Version control
    Authorization and approval
    Safe storage and retrieval
    Access control
    Regular review and updating

15. Key Points:
    Ensures qualified staff
    Tracks training history
    Supports audits
    Ensures regulatory compliance

16. **Measurement uncertainty due to calibration errors**
    Points:
    Incorrect calibration → bias
    Affects reliability
    Uncertainty propagation
    Need for regular calibration and traceability

17. **Role of internal audits**
    Points:
    Identifies deviations from SOPs
    Ensures corrective actions
    Promotes continuous improvement
    Supports accreditation requirements

18. Safety precautions for hazardous chemicals

Points:
Use of PPE
Proper labeling
Chemical storage rules
Handling with fume hoods
Emergency procedures

19. Purpose and functions of ISO

Points:
Develops international standards
Ensures quality, safety, efficiency
Provides uniform guidelines
Supports global harmonization

PART C – Essay

20. Standard safety precautions & risk assessment for hazardous chemicals

Expected points (any 8–10):
Identification of hazards
Risk assessment steps (likelihood, severity)
Use of MSDS
PPE requirement
Ventilation/fume hoods
Proper labeling
Chemical segregation
Storage of flammable/volatile substances
Emergency procedures
Spill management
Waste disposal guidelines

Explanation of at least 8 points may give full marks and 6–7 points → 8 marks, 4–5 points → 6 marks, 2–3 points → 4 marks

21. Expected coverage:

Traceability
SOPs
Record keeping
Audit trails
Error monitoring
Avoiding miscommunication
Consistency & repeatability
Compliance with ISO standards

Explanation of at least 8 points may give full marks and 6–7 points → 8 marks, 4–5 points → 6 marks, 2–3 points → 4 marks